

Reverse Engineering and Database of Off-the-Shelf Propellers for Middle-Size Multirotors

Yeong-Ki Jung*, Kyoungsik Chang^{*,§}, Sang-Hwan Park*, Van Thanh Ho*,
Ho-Joon Shim[†], Min-Woo Kim[‡]

^{*}*Department of Mechanical Engineering, University of Ulsan,
93 Daehak-ro, Nam-gu, Ulsan 44610, Republic of Korea*

[†]*Department of Unmanned and Autonomous Engineering,
Kyungwoon University, Gumi-si,
Gyeongbuk-do 39160 Republic of Korea*

[‡]*Unmanned Vehicle Advanced Research Center,
Korea Aerospace Research Institute Yuseong-gu,
Daejeon 34133, Republic of Korea*

In this work, 22 off-the-shelf propellers of 15–40 inches in diameter are reverse-engineered using a scanning system and reverse engineering software. Geometric information, including the diameter and pitch of the propeller and its airfoil sections, are extracted, and the chord length and twist angle of the extracted airfoil sections are investigated. All processes for reverse engineering are validated through comparison with design drawing information, product and performance data of the KPROP propeller designed by Korea Aerospace Research Institute. Based on the validated process, a database of 22 commercial propellers from four companies is constructed containing all the reverse-engineered information. The chord length, the maximum length and twist angle with respect to the diameter of the extracted airfoil section are analyzed in comparison with each other depending on the company and propeller size. In addition, the thrust and torque are simulated using computational fluid dynamic methodology, and the predictions are compared with reference data provided by the manufacturers.

Keywords: Multirotor; off-the-shelf propeller; reverse engineering; computational fluid dynamics; 3D scan.



1. Introduction

Recently, applications of multirotor drones have become widespread in various fields, such as agriculture, communication, rescue, surveillance, the military and meteorology. The size of the airframe tends to increase rapidly to extend the short flight endurance and time caused by the small

batteries used in multirotor systems. Multirotor transportation systems, which transport passengers or cargo without any human intervention, are a challenging future technology. They will require an increase in motor capacity and propeller size to generate the higher thrust and torque required for heavier payloads.

The aerodynamic and structural design of a propeller is important when designing large multirotors, such as for a transportation system. However, if the geometric information including the diameter and pitch of the propeller and its airfoil section of previously designed propellers are not provided, it is time-consuming and difficult to design a large, optimized propeller from scratch. Therefore, data about existing propellers, especially off-the-shelf ones, will be greatly helpful in designing new, large propellers. Many

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Email Address: kschang76@ulsan.ac.kr

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research groups have already used experiments and numerical simulations to investigate the geometry, performance and noise of off-the-shelf propellers for small multirotors.

The University of Illinois at Urbana-Champaign (UIUC) propeller dataset [1–3] provides wind tunnel measurements for the performance of commercial propellers used on small Unmanned Air Vehicle (UAV) and model aircraft. The UIUC research group considered 210 propellers (140 in volume 1 and 70 in volume 2 [1]) that they purchased off-the-shelf from retail outlets. The chord length and twist angles of each propeller, which ranged in size from 2.25 inches to 11 inches in diameter, were measured using PropellerScanner software by Hepperle [4]. The wind tunnel measurements include the thrust and torque coefficients in static conditions at various revolutions per minute (rpm) and over a range of advance ratios at the specified rpm. Kolaei et al. [5] also conducted wind tunnel experiments with T-motor 18 × 6.1-inch and Master Airscrew Electric 11 × 7-inch propellers with the rotor at various inflow angles. They presented propeller performance data such as thrust, power and roll moment coefficients for both propellers. Nowicki [6] studied the relationship between rotor blade deformation and rpm in hover mode for two commercial multirotor propellers: DJI Phantom 3 and T-motor 15 × 5-inch. He measured the out-of-plane deflection and change in pitch using a digital single-lens reflex camera and compared the results with those from a finite element analysis simulation.

In the process of design of the propeller or prediction of the aerodynamic performance, the scanning and reverse engineering of the existing propeller such as off-the-shelf or developed one also can be required to modify and optimize them. Based on the extracted geometric information, the aerodynamic performance is predicted using the method of blade element momentum theory (BEMT) or computational fluid dynamics (CFD) and validated through experiments [7–12]. MacNeill and Verstraete [7] considered five APC Thin Electric propellers (10–19 inches in diameter) and extracted airfoil section information using a David-3D SLS-2 3D scanner. They extended the generic BEMT with Viterna–Corrigan flat plate theory and Corrigan–Schillings stall delay model to acquire simulation results that closely followed

the experimental data. McCrink and Gregory [8] applied a BEM model with corrections for tip losses, Mach effects, three-dimensional flow components and Reynolds scaling to three different pitch propellers based on APC’s 10-inch-diameter propellers. The BEM modeling was validated with results from the wind tunnel tests, and they showed that the proposed BEM model was suitable for propeller design for UAV. Kutty and Rajendran [9] used a small propeller, APC 10 × 7-inch in size, to predict the thrust, power coefficients and efficiency through various advance ratios with CFD, and they compared their numerical predictions with experimental data.

In addition to the performance of multirotor propellers, the noise they generated has been investigated experimentally [13, 14]. Wisniewski et al. [13] measured thrust and sound pressure levels using the 9.4 inch diameter DJI stock propeller, 9.4-inch, and several custom-designed propellers. They found that the local sound pressure level was created by the local shear flow, and they presented an improved design that offered the required thrust while being 20 dB quieter. Brungart et al. [14] tested 13 off-the-shelf propellers with diameters from 6 inches to 24 inches to measure the noise in static thrust conditions in an aero-acoustic wind tunnel. They applied a simple scaling relationship based on the fan affinity law and achieved a significant noise reduction and efficiency increase through experiments measuring the radiated noise and power.

Most previous researches on off-the-shelf propellers for multirotors have focused on small propellers (11 inches in diameter or less), which have a low Reynolds number flow region. As the requirements for multirotor performance have increased, research on middle-size or large propellers has been reported. Thurlbeck and Cao [15] analyzed and modeled UAV power system architectures using T-motor 28- and 30-inch propellers. Their simulation showed that those commercial propellers could carry the heavier payload of a hybrid fuel cell stack and battery architecture or increase the endurance.

In this work, medium-size propellers (15–40 inches in diameter) are reverse-engineered to investigate their geometry and performance. As shown in Table 1, 22 propellers from five companies (T-motor, KDE, Dualsky, Tarot and XOAR) are considered. The geometric information for

Table 1. List of off-the-shelf propellers tested in this work.

inch	15	16	17	18	19	19	20	22	23	24	26	27	28	29	30	32	34	40	Index
T-Motor	●	●	●	●				●		●	●	●	●	●			●	●	T
Dualsky	●	●													●				Du
KDE	●			●						●									K
Tarot		●		●															Ta
XOAR	●			●															X

the propellers were scanned using an ATOS II Triple Scanner [16], and the surface and volume of the propellers were reverse-engineered using Geomagic Design X software [17]. The diameter and pitch of the propellers and airfoil section information (chord length, twist angle and thickness) were extracted from the corrected geometric models. Also, the thrust and torque are predicted numerically and compared with reference data provided on pertinent company's website. All processes, including scanning, reverse engineering and CFD, were validated with the KPROP propeller, for which the design and product drawings were provided.

2. Geometric Modeling and Reverse Engineering

To extract geometric information, each of the selected propellers was scanned on an ATOS II Triple Scanner, a noncontact optical laser scanning sensor developed by GOM company. The detailed specifications are shown in Table 2. To minimize the effect of surface reflectivity and extract the exact geometric information, a scan spray coating was placed on the propeller surface, and about 30 recognition points were attached to the propeller surface for position recognition. Each propeller was fixed at the clamp using the magnetic base on the base turntable, and the geometry was scanned five or ten times repeatably in one measurement, as shown in Fig. 1. The scanning sizes of adopted lenses are $100 \times 75 \times 70$ (mm³), MV100 for small propellers, and

$320 \times 240 \times 240$ (mm³), MV320 for large ones. In the case of MV100, the uncertainty of this scanning system is measured to be about 0.0006 mm in the standard deviation for scanning a 12-mm-diameter cylinder [18].

For propellers 34 inches or larger, the scanning instrument itself was rotated around the propeller model instead of rotating the turntable containing the model to prevent the blade tip of the propeller from vibrating during the scanning process due to its weight. That method requires much more time and expertise than rotating the turntable.

Because the raw data from 3D scanning generally includes position information for only the scanned geometry, the surfaces are not smooth, and it is difficult to obtain exact information about the surface and volume. Therefore, a correction process is required to calibrate the raw data based on the reference positions and make a smooth surface. This work used the Geomagic Design X [17] reverse engineering software to calibrate and correct the raw data. After the STL files of raw data were opened using Design X, the original models were modified and fixed. The overall process and functions adopted in this software [17] to correct the propeller geometry are listed as follows:

- (1) *Fill holes*: Fill missing holes with poly-faces based on the local mesh shape.
- (2) *Defeature*: Remove selected poly-faces and repair the area with intelligent hole filling.
- (3) *Healing wizard*: Automatically heal various defects in the mesh.
- (4) *Enhance shape*: Improve the quality of the mesh by sharpening corners and smoothing flat or rounded areas.
- (5) *Global remesh*: Recalculate the entire mesh and improve the mesh quality.

Finally, the surface and volume geometry were saved as STP files to enable the extraction of geometric information, such as the airfoil sections, chord lengths and twist angles of the propellers.

Figure 2 shows the raw and modified geometries before and after reverse engineering using Design X. As shown in Fig. 2(a), the raw geometry has some holes, nonsmooth surfaces on the blade and rough and blunt shapes near the trailing edge. After reverse engineering through the presented process, the holes are filled and the surface is smoothed, as shown in Fig. 2(b). In the propeller geometry, the trailing edge and the blade tip are too sharp to scan the exact geometry, even though the scanning system has high resolution. Therefore, discrepancy between the original and reverse-engineered geometry is unavoidable, as described below. At this point in our investigation, we did not apply any extra treatment to make the trailing edge or blade tip sharp but instead preserved as much of the original geometry in the scan as possible. The surface and volume

Table 2. Specifications of the ATOS II Triple Scanner.

Sensor	Value(s)
Camera pixels	$2 \times 5,000,000$
Measuring area	$38 \times 29\text{--}2000 \times 1500$ mm ²
Point spacing	0.02–0.79 mm
Working distance	490–2000 mm
Measured point per scan	5 million points
Operating temperature	5–40°C



Fig. 1. ATOS II Triple Scanner by GOM [16] and the scanning process.

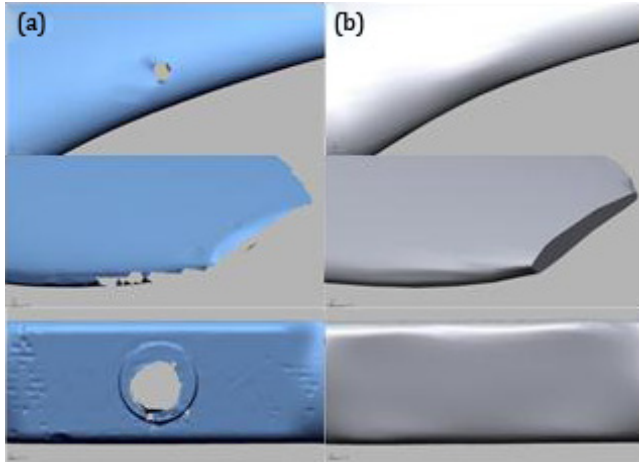


Fig. 2. Geometries (a) before and (b) after reverse engineering using Design X.

information generated through reverse engineering and stored in STP files was used to extract propeller information such as the pitch and airfoil section. The domain of the half propeller was divided into ten sections from the hub to the blade tip, and each airfoil geometry was extracted using CATIA V5 design software [19]. The chord length and twist angle of the extracted airfoil sections were measured as shown in Fig. 3. The nondimensional airfoil section can be

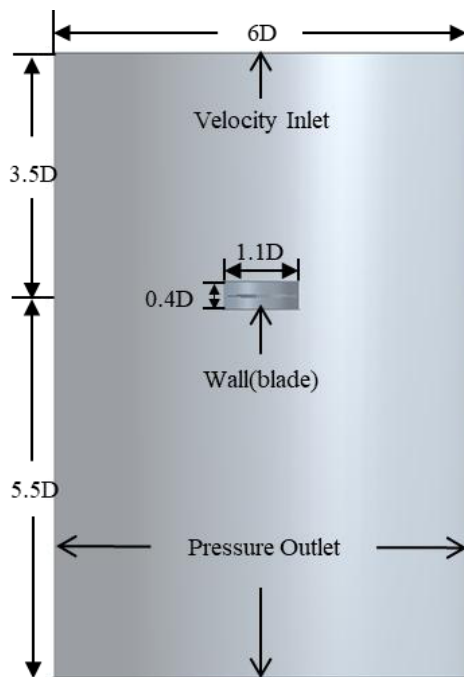


Fig. 3. The process of extracting airfoil section information: (a) division into ten sections, (b) twist angle and chord length, (c) nondimensional airfoil section, (d) reference point for pitch and (e) twist angle.

obtained by rotating the measured twist angle. The pitch is calculated using the following formula (1) based on the twist angle (θ) at the 75% radius ($0.75R$) point of each propeller:

$$\text{Pitch angle}(\beta) = 2\pi \times \tan \theta \times 0.75R. \quad (1)$$

3. Numerical Simulation of Propeller Performance

Based on the extracted propeller geometry, we used the CFD method to predict the thrust, power and torque in hovering flight conditions. In this work, ANSYS Fluent V19 commercial software [20] was used to generate the mesh and solve the fluid dynamics governing equations (incompressible Navier–Stokes). The pressure-based solver was set as the solver type, and the coupled scheme for pressure–velocity coupling was selected as the solution method. The gradient terms in the governing equations are discretized by the least square cell-based method, and the remaining terms in the momentum and turbulence equations are discretized in the second-order upwind scheme. The

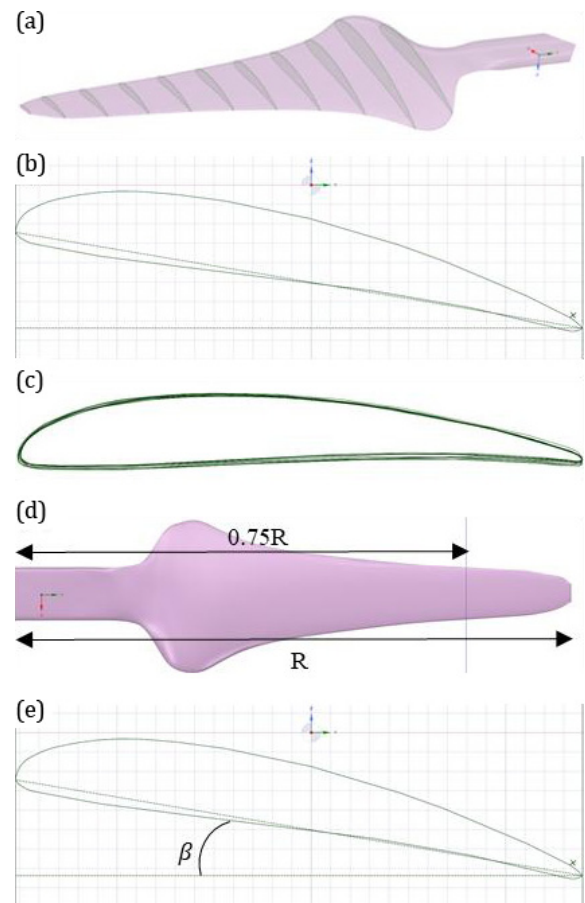


Fig. 4. Computational domain and boundary conditions for numerical simulation in hovering flight conditions.

pseudo-transient solution method is used with implicit under-relaxation to obtain faster and more robust solutions. The turbulent eddy viscosity is calculated from the SST $k-\omega$ turbulence model. To simulate the rotating propeller in hovering flight conditions, the multiple reference frame (MRF) model, which considers the absolute velocity formulation and an extra source term related to additional acceleration, is adopted.

The velocity inlet boundary condition with zero velocity (hovering flight condition) is used for the inlet of the computational domain, and the pressure outlet boundary condition is set at the outlet and far-field. The no-slip boundary condition is applied to the surface wall of the propeller. The computational domain consists of a cylinder type with six times propeller diameter ($6D$) and nine times propeller diameter ($9D$) as the diameter and height, respectively, where D is the diameter of the propeller. The heights of the freestream and backflow parts of the propeller are $3.5D$ and $5.5D$, respectively, as shown in Fig. 4. To apply the MRF method, the computational domain is divided into two regions: inner and outer regions. The size of the inner region is set to $1.1D$ diameter and $0.4D$ height. The outer region is set to $6D$ diameter and $9D$ height with $3.5D$ towards the inlet and $5.5D$ to the bottom outlet. This work does not consider the adopters and motors assembled with the propeller to minimize the simulation cost and maximize efficiency. Only the propeller in the hovering flight condition is simulated numerically with variations in the rpm.

4. Validation

We validated the overall process, including the scanning, reverse engineering and extraction of geometric information, using a 22-inch-diameter KPROP propeller developed by the Korea Aerospace Research Institute (KARI) [21], which provides product and design drawings, as the reference. The geometry of the scanned and reverse-engineered model is compared with the design drawing. The details of the airfoil geometry are also validated using information in the design drawing. To evaluate the adequacy of the adopted CFD method and its numerical schemes and mesh, the simulation results are compared with experimental data and extra simulation results for the original model. There could be any level of tolerance between the design drawing and the model produced in the manufacturing stage. However, this work neglects that tolerance, assuming that the produced model is nearly the same as the design drawing.

Table 3 shows the differences in the propeller geometry, calculated using the CATIA V5 design software [19], between the original KPROP [21] propeller model and the reverse-engineered one. The length and surface area of both

Table 3. Comparison of geometric information between the original and reverse-engineered KPROP propeller models.

KPROP	Diameter (mm)	Area (mm ²)	Volume (mm ³)
Reference	558.80	47,108.49	92,405.06
Model	557.90	46,857.52	98,604.91
Error (%)	−0.16	−0.53	6.71

models differ by less than 1%, and the entire volumes of the original and reverse-engineered models are 92,405.06 mm³ and 98,604.91 mm³, respectively, for a 6.71% difference.

Even though the length and area of the reverse-engineered model are smaller than the original model, the volume of the reverse-engineered model is much larger because the airfoil sections, including the trailing edge, were scanned as thicker than the original, as shown in Fig. 5. The left-hand side of the figure corresponds to the original model, and the right-hand side shows the reverse-engineered model. As expected, even though the overall geometry looks the same, a small discrepancy occurs near the trailing edge, with the reverse-engineered model showing a blunter trailing edge than the original model due to the resolution limit of the scanning system. The discrepancy is larger in station 10 than in station 2 as it moves from the hub to the blade tip.

Among the ten stations, the chord length and twist angle show the maximum values near station 2, though this does not mean that the two variables are maximum exactly at station 2 over the entire propeller, and they decrease with the distance from the hub to the blade tip. The chord length of each airfoil section in the reverse-engineered model agrees well with that in the original design model, as shown in Fig. 6. The difference between them is within 3%, with the reverse-engineered model's chord lengths a little bit too

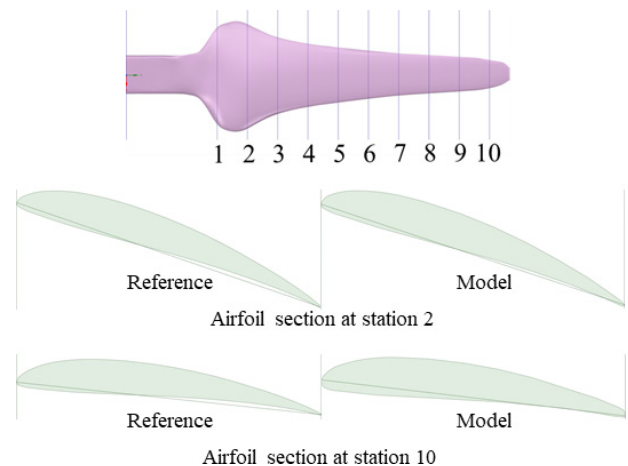


Fig. 5. Comparison of airfoil sections between the original and reverse-engineered models.

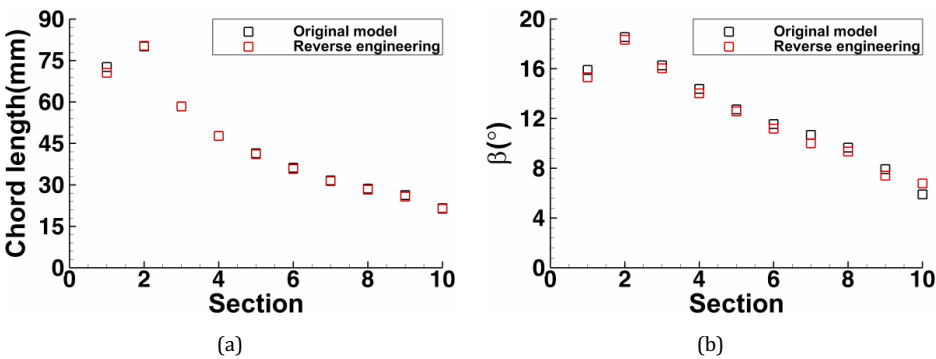


Fig. 6. Comparison of chord length and twist angle in each airfoil section between the original and reverse-engineered models.

Table 4. Comparison of chord length and twist angle in each airfoil section between the original and reverse-engineered models.

Chord length (mm)				Twist angle β (degrees)			
Original model	Reverse-engineered model	Error (%)		Original model	Reverse-engineered model	Error (%)	
1	72.65 mm	70.54 mm	2.91	1	15.90°	15.31°	3.71
2	80.08 mm	80.31 mm	0.28	2	18.55°	18.34°	1.13
3	58.38 mm	58.32 mm	0.09	3	16.28°	16.03°	1.54
4	47.73 mm	47.68 mm	0.12	4	14.38°	14.02°	2.50
5	41.43 mm	41.12 mm	0.75	5	12.73°	12.57°	1.26
6	36.18 mm	35.75 mm	1.18	6	11.55°	11.18°	3.20
7	31.59 mm	31.32 mm	0.86	7	10.68°	10.00°	6.37
8	28.63 mm	28.28 mm	1.23	8	9.66°	9.34°	3.31
9	26.29 mm	25.76 mm	2.01	9	7.93°	7.40°	6.68
10	21.56 mm	21.33 mm	1.03	10	5.90°	6.78°	14.92

small at every station (Table 4). The twist angle increases up to the second station and then decreases smoothly until the blade tip (Fig. 6). The range of this angle is from 5° to 20°. In the comparison, the twist angle differs by less than 5% in nine stations; station 10, the blade tip, has a 14.9% difference in the twist angle between the original and reverse-engineered propellers. That big difference near the blade tip occurs because the propeller is thinnest there and has a very sharp trailing edge.

This validation confirms that the present procedures for scanning, reverse engineering and extracting the airfoil information are appropriate given the measurement accuracy of other reverse engineering studies [10, 22] for drone or aircraft propellers. Therefore, we applied this process to conduct reverse engineering of 22 commercial propellers with diameters of 15–40 inches and compiled a database of their chord lengths, twist angles and airfoil geometries.

The performance of the reverse-engineered KPROP propeller was validated through comparison with experimental data [21] and the simulation result of the original model KPROP. Before presenting that performance comparison, we report on a grid test conducted to find an appropriate mesh for conditions at 3000 rpm. Three types of

mesh, coarse (2.8×10^6 cells), medium (5.1×10^6) and fine (7.3×10^6), are considered, and the calculated thrust coefficients (C_t) are 0.0746 (coarse), 0.0756 (medium) and 0.0766 (fine), as shown in Table 5. The grid test indicates that the medium mesh offers the best computational cost and time for this work. As shown by the contours of the wall unit, y^+ on the front and back sides of the propeller, all regions have a wall unit of less than $y^+ \approx 1$, which means that our grid system is appropriate for use with the SST $k-\omega$ turbulence model. Figure 7 shows the values of thrust (N) and torque (N · m) of the original model, the reverse-engineered model and the experimental measurements [21] with respect to 1000–4000 rpm.

Table 5. Grid test of reverse-engineered KPROP propeller.

Mesh	No. of cells	Thrust (N)	C_t	Experiment [17]	
				Thrust (N)	C_t
Coarse	2,796,362	22.32	0.0746	24.31	0.0814
Medium	5,107,952	22.61	0.0756		
Fine	7,301,194	22.91	0.0766		

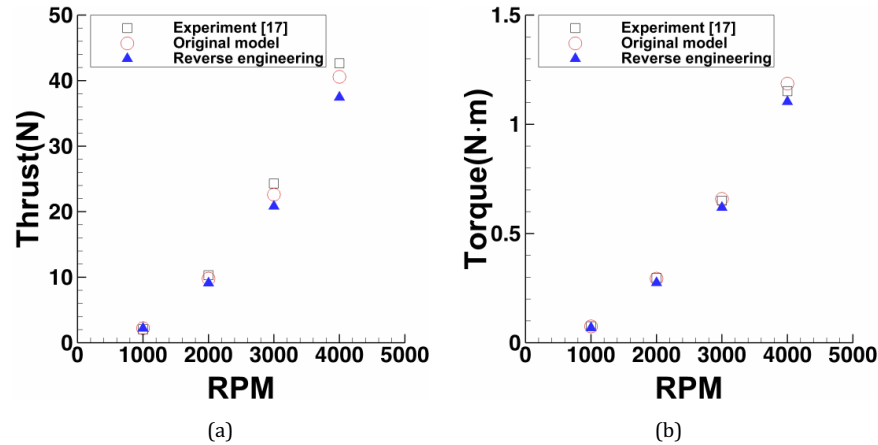


Fig. 7. Comparison between experimental data, original and reverse-engineered KPROP propeller models: (a) thrust and (b) torque.

Overall, the results of the reverse-engineered model agree well with the results of the original model and the experimental measurements when the differences between the original model and the experimental values are considered. However, the differences among the experimental data, original model and reverse-engineered model also increase as the rpm increases. The predicted thrust of the reverse-engineered model has the lowest value among the three cases. The thrust and torque of the original and reverse-engineered models at the condition of 3000 rpm differ from each other by 7.9% and 6%, respectively.

5. Results

We analyzed the specifications of the off-the-shelf propellers and the extracted airfoil sections using reverse-engineered models of propellers from five companies [23–27]. The aerodynamic performance (thrust and torque)

is predicted by comparing the results from the CFD method with performance data sheets on the company website [23].

The diameter and pitch of the reverse-engineered models are compared with the specifications on the company websites in Fig. 8. The X- and Y-axes correspond to the specifications from the company and the values measured in the reverse-engineered models, respectively. The measured diameters of the propellers agree well with the reference data, with an average difference of 1%. The average difference in pitch between the reverse-engineered model and the reference data is about 8.4% in all models. As the diameter of the propeller increases, the difference in pitch between the specifications and the models decreases, as shown in Fig. 8. This indicates that the scanning system had difficulty in exactly capturing the geometry of small propellers. The measured pitches of the T-motor and Dualsky models show the smallest differences, of about 4.6% and 7%, respectively, from the reference values, and the XOAR propellers had the biggest difference (23.6%).

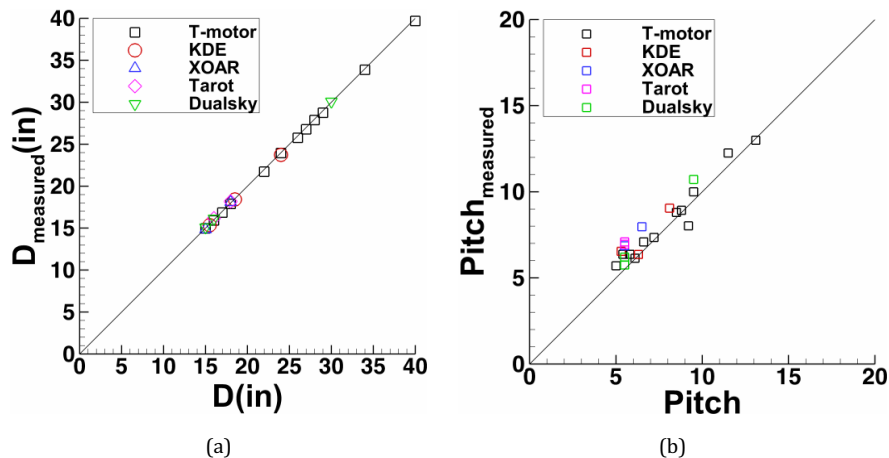


Fig. 8. Comparison between model specifications and the reverse-engineered values: (a) diameter and (b) pitch.

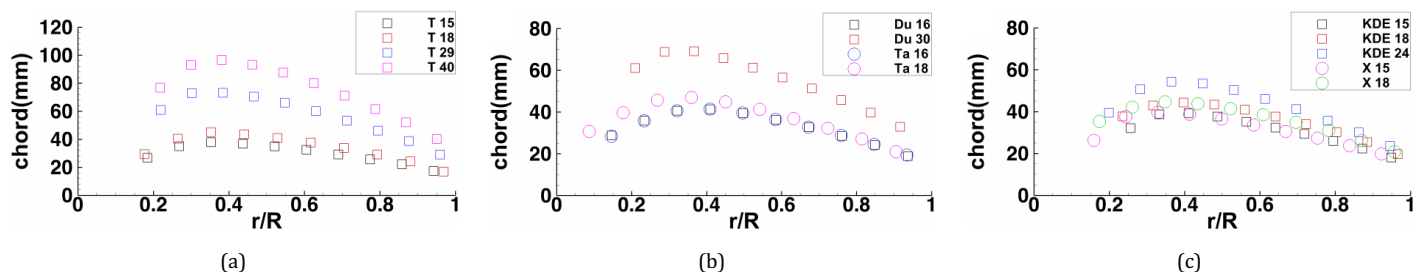


Fig. 9. Chord length (mm) with respect to the radius of the propeller: (a) T-motor; (b) Dualsky and Tarot; and (c) KDE and XOAR.

The chord length, with respect to the propeller radius, is plotted in Fig. 9 according to the nondimensional propeller radius (r/R).

As the distance from the hub of the propeller increases, the chord length of the airfoil section increases up to about $0.4r/R$ and then decreases smoothly in all models. The decay rate of the chord length after the maximum point depends on the size of the propeller and the company. In Brandt and Selig's [2] study of small propellers, the APC (11×8 -inch) and Graupner (11×8 -inch) models also have a maximum chord length of about $0.4r/R$, with a gradual decrease after this position until $0.05c/R$ or below at the blade tip. In the case of propellers with a 15-inch diameter, it is interesting that the variation of chord length in the radius is consistent in all models measured in this work. The maximum chord length of all scanned propellers was investigated and plotted with an approximate regression line in Fig. 10. The X- and Y-axes correspond to the diameter and maximum chord length of the propellers in inches, respectively. The formula of the regression line is estimated to be $0.0932x + 0.0952$ with an R^2 value of 0.98.

Figure 11 shows the variations in the twist angle in all models, with respect to r/R by degree. Most models have a similar variation of the twist angle: Independent of the

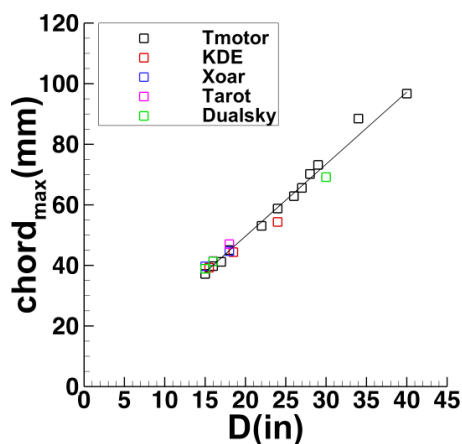


Fig. 10. Maximum chord length according to the diameter, regression line.

propeller size, the angle increases up to about $0.2r/R$, with a maximum value between 20° and 25° , and then decreases to near 5° . The station with the maximum twist angle is between $r/R = 0.2$ and $r/R = 0.25$. All the propellers from a single company have nearly the same variation in the twist angles. Brandt *et al.* [1] show that small propellers from most companies have the maximum twist angle at $r/R = 0.2$, which is similar to what we found with middle-size propellers in this work. However, in the small propellers, the maximum twist angle varies from 25° to 55° , and the twist angle at the blade tip is from 10° to 15° . For example, the APC 11×8 -inch and Master Airscrew 11×8 -inch propellers have the maximum twist angles of about 52° and 27° , respectively.

The airfoil sections, which are nondimensionalized by the corresponding chord length, are shown in Fig. 12. In this paper, airfoil information is presented only for selected models because of space limitations. The dataset of all models is available in our website or through e-mail. The thickness of the airfoil section is greatest near the hub and becomes thinner or remains constant with an increase in r/R .

In their investigation of small propellers, Deters *et al.* [3] found the same tendency in NR640 propellers (5 inches and 9 inches), which are large near the hub and become small as the radius increases. The variations in the thickness and shape of the airfoil section are large in small propellers and decrease as the size of the propeller increases. Therefore, the number of airfoil shapes adopted in all the propeller regions decreases as the propeller size increases. The thickness of the trailing edge tends to become large as the propeller size decreases. However, this tendency seems to be due to the resolution limit of the scanning system, which cannot capture the exact thickness of a small propeller. That problem disappears as the propeller size increases. The shapes of the airfoil section differ by company, as shown in Figs. 12(a) and 12(b). Figure 12(a) compares the airfoil sections of 18-inch propellers made by different companies. The T-motor and Tarot propeller models have maximum thicknesses of the airfoil near $x/R = 0.4$ in all sizes. However, the KDE models show a change in the maximum thickness from $x/R = 0.2$

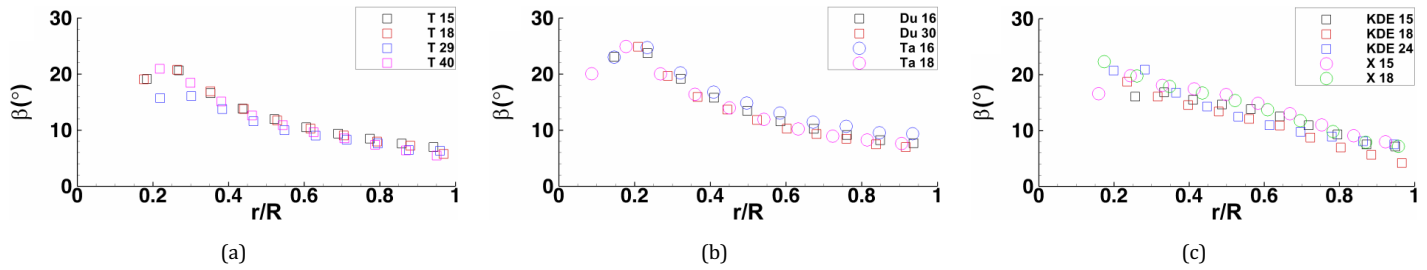


Fig. 11. Twist angle (degree) with respect to the radius of the propeller: (a) T-motor; (b) Dualsky and Tarot; and (c) KDE and XOAR.

to $x/R = 0.4$ in the 18- and 24-inch sizes. Figure 12(c) compares the airfoil sections of 24- and 30-inch (29-inch) propellers. At both sizes, the T-motor models have a thinner airfoil than the Dualsky and KDE models.

The Dualsky models have the maximum thickness near $x/R = 0.3$ except for some sections near the hub. When the error between the original model and reverse-engineered model is considered, pointing out the exact position of

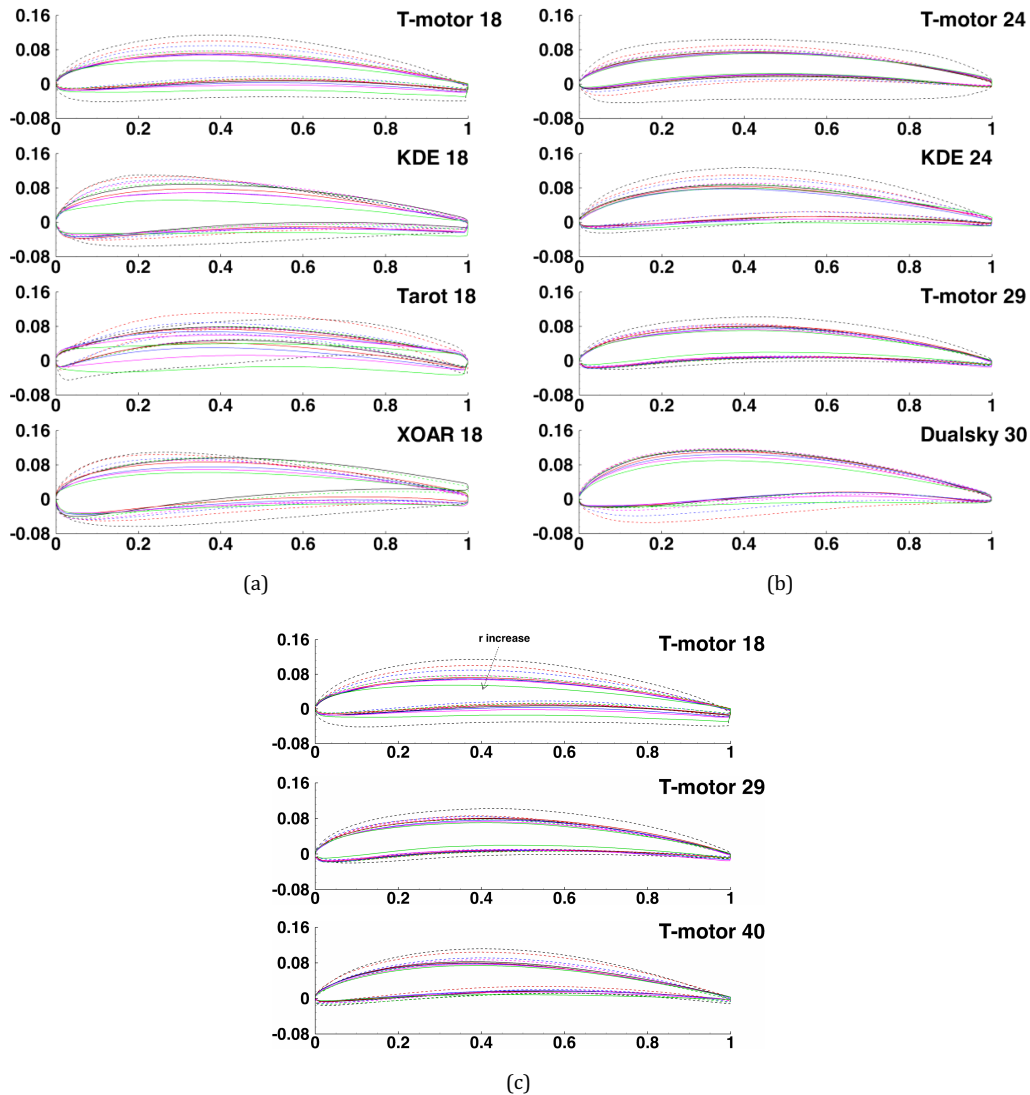


Fig. 12. Nondimensionalized airfoil section of each model (X -axis: x/c , Y -axis: y/c): (a) 18 inches (four companies), (b) 24 inches and 29 inches (30 inches) (three companies) and (c) T-motor (18, 29 and 40 inches).

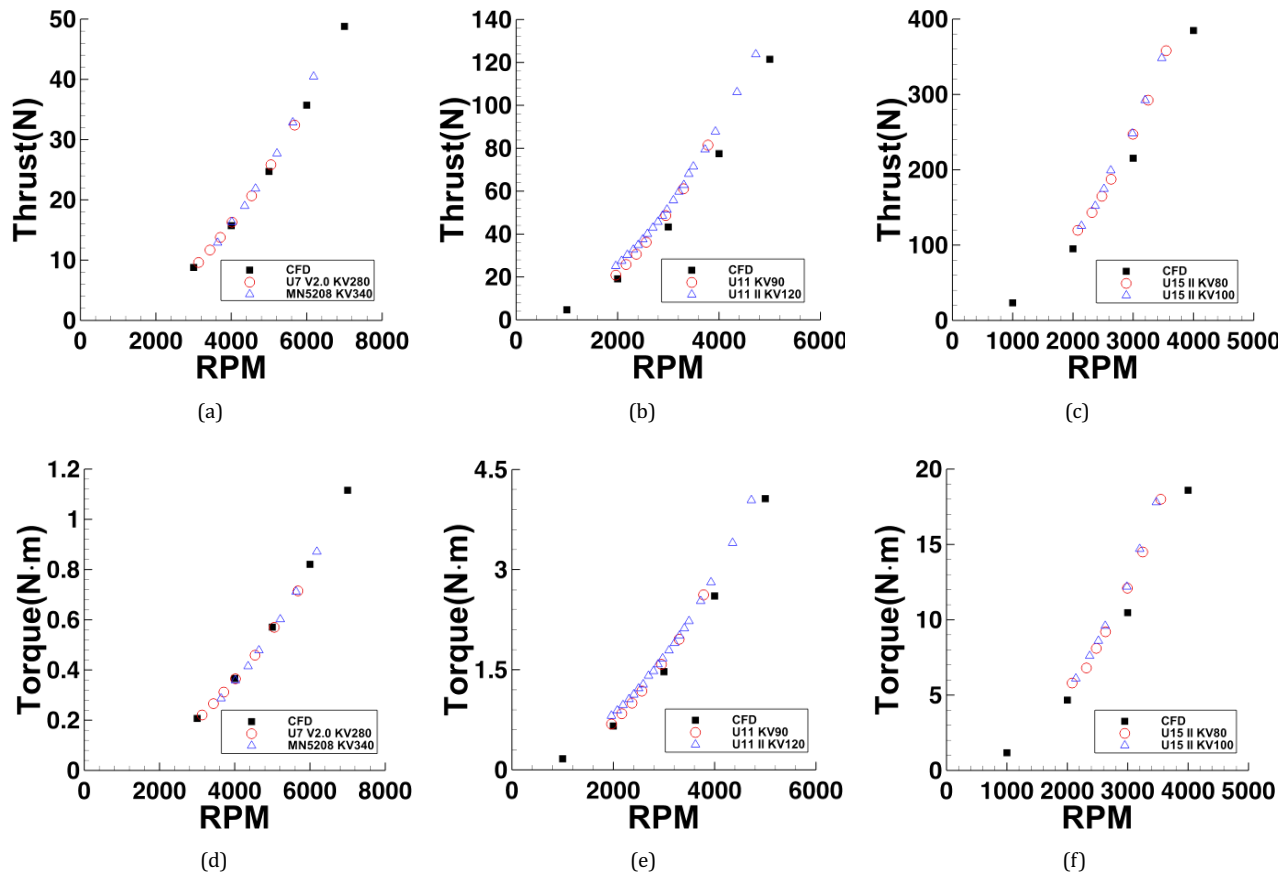


Fig. 13. Thrust (upper) and torque (lower) with respect to rpm for T-motor propellers of various diameters: (a), (d) 18 inches, (b), (e) 27 inches and (c), (f) 40 inches.

maximum thickness is not meaningful. Instead, we provide a rough estimation of the maximum position. All models except those from Tarot show that the thickness and the chamber line of the airfoil become thin as the distance from the hub increases. We did not find a consistent airfoil shape in the Tarot propellers.

Figure 13 compares the thrust and torque performances of 18-, 27- and 40-inch-diameter propellers from T-motor in hovering flight conditions. The simulation results are presented only for the T-motor propellers because the other companies do not provide performance data on their websites for comparison. The objective of the present simulation was not to evaluate the performance of the propeller but to evaluate the quality of our reverse-engineered models. The rectangular symbols correspond to simulation results using CFD, and the other symbols are reference data from the company [23]. Because T-motor provides reference data based on experimental results from a static thrust test with a combination of motor and propellers, there is a small difference in the performance depending on the choice of motor. The rpm range is 3000–7000 for the

18-inch propeller and is 1000–4000 for the 40-inch propeller. The predicted thrust and torque agree well with the reference data [23]. In particular, the agreement between the simulation results and the experiment is best with the 18-inch propeller. The simulation results for the larger propellers predict slightly lower values for the thrust and torque than in the reference material.

Even though the error of geometric measurement decreases as the propeller size increases as mentioned in previous section, the increase of the discrepancy between simulation results and experiment data can be caused by the deficiency in present CFD methodologies such as insufficient computational domain size, no consideration of compressibility effect and deflection of the propeller. The prediction of a low thrust coefficient by CFD compared with experimental data is consistent with previous numerical simulations [9,21]. As the rpm increases, the discrepancy between the CFD predictions and experimental data increases. When the validation results with KPROP propeller (22-inch) given above are considered, the under-prediction of thrust and torque is reasonable.

6. Conclusion

In this work, 22 off-the-shelf propellers (15–40 inches in diameter) are reverse-engineered through a scanning system, ATOS II Triple Scan, and reverse engineering software, Design X. Geometric information, diameter and pitch of the propeller and its airfoil sections, are extracted using CATIA V5 software. The chord length and twist angle of each airfoil section are also investigated. All processes are validated using the KPROP propeller by KARI as the reference propeller to compare the reverse-engineered model with the original model. The length and surface between the two models differ by less than 1%. The extracted airfoil section is modeled thicker than the original one, especially in the trailing edge.

This discrepancy, which becomes inevitable in reverse engineering process, is caused by the limitation of the scanning system resolution. The deterioration of the measurement quality can reduce the accuracy of numerical prediction when it is considered that the prediction of aerodynamic performance of an airfoil or propeller is very sensitive to the geometry such as the trailing edge. The chord length of the reverse-engineered model agrees well with the original one (within 3%), but the twist angle differs by 7% in most regions and 14.9% in the blade tip region. The thrust and torque of the reverse-engineered model were validated through the CFD methodology and differ from the original model by 7.9% and 6%, respectively. Using that validated process, a database of 22 commercial propellers was constructed. The measured diameter and pitch of all propellers differ by 1% and 8.4%, respectively, on average. The detailed analysis of geometry information such as chord length and twist angle of commercial middle-size multirotors is performed and can be used as reference data for the design and optimization of the propellers of this size. The variation in the chord length with respect to the distance from the hub is similar in all propellers, independent of the manufacturer. The maximum chord length of all models is estimated with a regression line. The twist angle variation of all models seems to be independent of propeller size.

The shape of the airfoil section differs by propeller size and manufacturer. The prediction of the thrust and torque of the propellers from T-motor agrees well with reference data from the company. As the propeller size and rpm increase, the discrepancy between our model and the reference data increases.

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Yeong-Ki Jung is currently doing his M.S. degree at the School of Mechanical Engineering, University of Ulsan, Republic of Korea. He received his B.S. degree from the University of Ulsan in 2019. His current research interests include aerodynamics of multirotors and aircrafts, compressible flow simulation, multiphase flow on a superhydrophobic surface and uncertainty quantification in CFD.



Kyoungsik Chang is an Associate Professor at the School of Mechanical Engineering, University of Ulsan, Republic of Korea. He received his B.S., M.S. and Ph.D. degrees from KAIST (Korea Advanced Institute of Science and Technology), Daejeon, in 2000, 2002 and 2007, respectively. His current research interests include aerodynamics of multirotors, turbulent flow simulation and uncertainty quantification in CFD.



Sang-Hwan Park is a currently doing his M.S. degree at the School of Mechanical Engineering, University of Ulsan, Republic of Korea. He received his B.S. degree from the University of Ulsan in 2020. His current research interests include reverse engineering, heat transfer simulation and PIV experiment.



Van Thanh Ho is a Researcher at the School of Mechanical Engineering, Chung-Ang University, Seoul. He obtained his B.S. degree in Engineering Mechanics from HCMC University of Technology, Vietnam, in 2017, and M.S. degree in Mechanical and Automotive Engineering from the University of Ulsan, Republic of Korea, in 2020. He works on negative isolation space, battery and artificial muscle networks in CFD.



Ho-Joon Shim is an Assistant Professor at the Department of Unmanned and Autonomous Vehicle Engineering, Kyungwoon University, Republic of Korea. He received his B.S. and M.S. degrees from the University of Ulsan, Republic of Korea, in 2006 and 2008, respectively, and his Ph.D. degree from KAIST (Korea Advanced Institute of Science and Technology), Daejeon, in 2015. His current research interests include aerodynamics of multirotors, multicopter flight test and wind tunnel experiments.



Min-Woo Kim is a Senior Research Engineer at the Korea Aerospace Research Institute, Daejeon. He received his B.S., M.S. and Ph.D. degrees from Seoul National University, Republic of Korea, in 2004, 2006 and 2012, respectively. His current research interests include vehicle design of unmanned systems and computational aeroacoustic simulation.